

A SUFFICIENT STATISTIC FOR DEPTH-ONE QUANTUM APPROXIMATE OPTIMIZATION

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Abstract. A growing literature learns priors over quantum approximate optimization algorithm (QAOA) angles, or transfers angles between graphs, in order to spend fewer measurements of the quantum objective. We show that at depth one on unweighted MaxCut this effort is unnecessary, and that its standard evaluation protocol is unsound. From the closed form of Wang, Hadfield, Jiang and Rieffel we prove that the depth-one landscape is a functional of a single probability measure carrying three integers per edge: the two endpoint degrees and the number of common neighbours. That measure is a sufficient statistic. Graphs that share it have identical landscapes even when they are non-isomorphic, of different order, and of different girth; we confirm this to machine precision for the Petersen graph and the pentagonal prism. The statistic yields a transfer guarantee that needs no quantum resources: two landscapes differ nowhere by more than the total variation distance between their measures, and transferring optimized angles costs at most twice that distance. We verify the guarantee on 186,624 graph pairs with 0 violations. Because the measure is computable in near-linear time, exact landscapes follow for graphs with more than a million edges, with no quantum queries, dominating every learned prior we tested. The statistic also exposes severe degeneracy in standard benchmarks: 96.6% of random cubic graphs share a landscape with another sample, and a random train-test split leaves 51% of test graphs with an exact landscape duplicate in training, inflating the apparent accuracy of a singular-value prior by an order of magnitude.

1. Introduction. The quantum approximate optimization algorithm [2, 4] prepares a variational state from a cost operator and a mixing operator and is tuned by measuring its objective on a quantum device. Each measured angle pair costs samples, and at shallow depth the number of angle pairs needed to locate a good setting dominates the experimental budget. This has motivated a substantial effort to guess good angles in advance: to transfer optimized angles between graphs [3, 5], to exploit the observation that the objective concentrates over instance ensembles [1], and to learn priors that rank angle grids before any target measurement is taken. The value of such methods is measured in queries saved.

At depth one on unweighted MaxCut, the objective has a closed form in local graph data. Wang, Hadfield, Jiang and Rieffel [6] showed that each edge contributes a term determined by the degrees of its endpoints and the number of vertices adjacent to both. The identity is established, and we claim no part of it. What has not been drawn out, and what this paper is about, is its statistical structure.

Written as a sum over edges, the identity says that the whole landscape is an average of one fixed function over the edges of the graph. Only the *distribution* of edge data enters, not which edge carries which value, and not any other property of the graph. The depth-one landscape is therefore a functional of a probability measure on three non-negative integers. We call that measure the edge statistic and show it is sufficient in the precise sense that it determines the landscape exactly.

Four consequences follow, and they are the substance of this work. First, the statistic is a strictly coarser invariant than isomorphism, so graphs with no structural resemblance can have numerically identical landscapes. Second, because the landscape is an expectation of a function bounded in $[0, 1]$, two landscapes are separated nowhere by more than the total variation distance between their statistics; this converts parameter transfer from an empirical practice into a certified one, computable classically in near-linear time. Third, the standard benchmark families of the QAOA literature are severely degenerate under this statistic, so a suite of random regular graphs contains far fewer distinct depth-one landscapes than it contains graphs. Fourth, since the statistic is cheap, the exact landscape is free, and no learned prior can beat zero queries; worse, the random train-test splits used to validate such priors place exact landscape duplicates on both sides and inflate their reported accuracy.

We report each of these with independent numerical verification, and we state plainly what the result does not cover. Everything here is depth one, unweighted and noiseless. Depth two and above, weighted objectives, and hardware sampling are outside the scope of the argument, and we do not extrapolate to them.

2. Results.

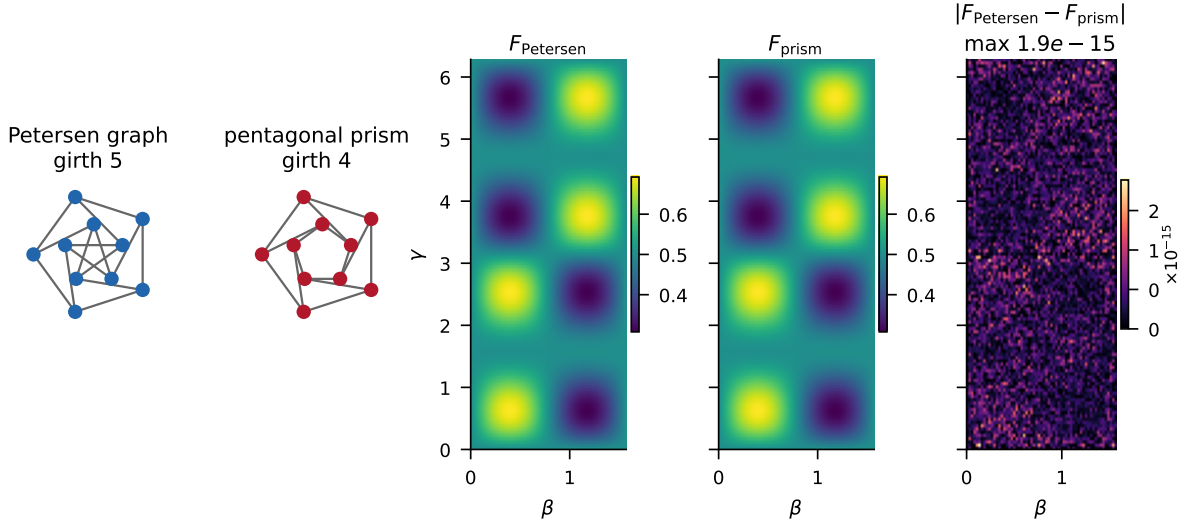


Fig. 1: Two graphs with nothing in common have the same depth-one landscape. The Petersen graph and the pentagonal prism are non-isomorphic and have different girth (5 and 4), but identical edge statistics. Their landscapes, each computed by an independent statevector simulation rather than by the formula, agree everywhere to 1.89×10^{-15} , which is roundoff. The right panel shows the difference on a scale of 10^{-15} .

2.1. The landscape is a functional of an edge statistic. For an unweighted graph $G = (V, E)$ with $n = |V|$ and $m = |E|$, write $C = \sum_{(u,v) \in E} (I - Z_u Z_v)/2$ and $B = \sum_{v \in V} X_v$, and let

$$F_G(\gamma, \beta) = \frac{1}{m} \langle + | e^{i\gamma C} e^{i\beta B} C e^{-i\beta B} e^{-i\gamma C} | + \rangle \quad (2.1)$$

be the normalized depth-one objective, with $|+\rangle$ the uniform product state. For an edge (u, v) define the triple

$$\tau(u, v) = (\deg u - 1, \deg v - 1, |N(u) \cap N(v)|) \quad (2.2)$$

and let μ_G be the empirical distribution of τ over the m edges, a probability measure on $\mathbb{Z}_{\geq 0}^3$. We call μ_G the *edge statistic*. Proposition 3 in the Methods restates the identity of Ref. [6]: the contribution of edge (u, v) is a fixed function $f(\tau(u, v); \gamma, \beta)$ of its triple alone. Summing and dividing by m gives the following.

Theorem 1 (Sufficiency). *For every graph G and every angle pair,*

$$F_G(\gamma, \beta) = \mathbb{E}_{\tau \sim \mu_G} [f(\tau; \gamma, \beta)]. \quad (2.3)$$

Consequently $\mu_G = \mu_H$ implies $F_G \equiv F_H$, whether or not G and H are isomorphic and whether or not they have the same number of vertices.

Equation (2.3) is an immediate consequence of the identity of Ref. [6], and we present it as such. Its content is the converse reading: everything the depth-one landscape can express about a graph is already contained in three integers per edge, discarded down to their distribution.

2.2. Landscape collisions between unrelated graphs. Theorem 1 predicts exact collisions, and they occur among familiar graphs. The Petersen graph and the pentagonal prism are both cubic on ten vertices, are not isomorphic, and have girth 5 and 4 respectively; every edge of each has triple $(2, 2, 0)$, so their statistics coincide. Figure 1 shows their landscapes computed by two independent statevector simulations, which agree to 1.89×10^{-15} . The Heawood graph and the seven-prism collide in the same way, to 7.77×10^{-16} . Collisions also cross graph sizes: the 6-cycle and the 8-cycle have identical landscapes to 6.66×10^{-16} , since both consist entirely of triples $(1, 1, 0)$.

Collisions are not generic. Exhaustive enumeration of connected graphs finds none on five or six vertices, and only 1.2% of the 853 connected graphs on seven vertices share a landscape with another graph of that order. The picture changes completely on structured families, which is where the consequence bites.

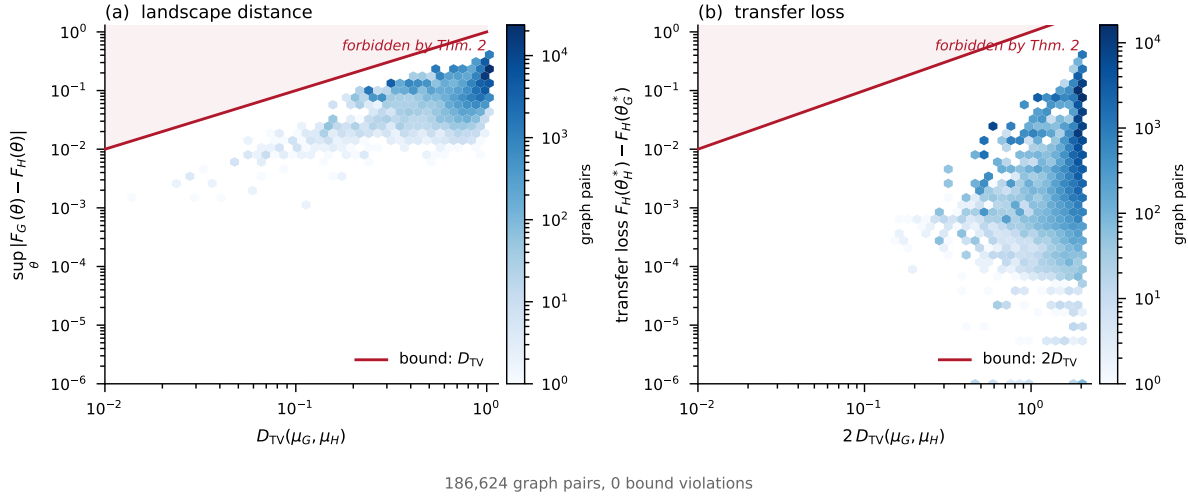


Fig. 2: The transfer certificate holds everywhere. Density of 186,624 graph pairs. **a**, Realized landscape separation against the total variation distance between edge statistics, with the bound of Eq. (2.4). **b**, Realized loss from transferring optimized angles, against the bound of Eq. (2.5). The shaded region is excluded by Theorem 2; no pair enters it.

2.3. A transfer guarantee that needs no quantum resources. Because $C_{uv} = (I - Z_u Z_v)/2$ is a projector, f takes values in $[0, 1]$, and Eq. (2.3) is an expectation of a bounded function. Comparing two graphs then reduces to comparing two measures.

Theorem 2 (Certified transfer). *For all graphs G, H and all angle pairs θ ,*

$$|F_G(\theta) - F_H(\theta)| \leq D_{\text{TV}}(\mu_G, \mu_H). \quad (2.4)$$

If moreover θ_G^ and θ_H^* maximize F_G and F_H over a common set of angles, then transferring the angles optimized on G to H costs*

$$F_H(\theta_H^*) - F_H(\theta_G^*) \leq 2 D_{\text{TV}}(\mu_G, \mu_H). \quad (2.5)$$

Both bounds are uniform in the angles and involve no quantum quantity. The right-hand side is a distance between two histograms of integer triples, computable from the two graphs alone in near-linear time. Parameter transfer at depth one therefore comes with an a priori certificate: before running anything, one can bound exactly how much is lost by reusing another graph’s angles.

We tested the bounds on a corpus of 432 graphs comprising breadth-first neighbourhoods of unweighted Stanford Gset instances [8] together with Erdős–Rényi, random regular and Barabási–Albert samples, giving 186,624 ordered pairs. Landscapes were evaluated at 4,608 angle points. Figure 2 plots the realized landscape separation and the realized transfer loss against their certificates. There are 0 violations. The bounds are valid but not tight: the median ratio of realized separation to certificate is 0.21, and the rank correlation between them is 0.62.

2.4. Standard benchmark families are degenerate. Random regular graphs are the default benchmark of the QAOA literature [7, 9]. They are also, at depth one, close to a single instance. Every triangle-free cubic graph has all edges of type (2, 2, 0) and therefore exactly one landscape, whatever its size or structure. Sampling 560 random cubic graphs on up to thirty vertices yields only 67 distinct depth-one landscapes, and 96.6% of the sample shares a landscape with another member. Higher degree helps but does not resolve the problem: 520 random quartic graphs give 186 landscapes with 84.4% sharing, and 520 random quintic graphs give 349 landscapes with 48.1% sharing (Fig. 3).

This is a statement about the benchmark, not about the algorithm. A study that reports depth-one performance averaged over a random regular ensemble is averaging over far fewer distinct problems than it appears to be, and its error bars do not carry the independence they assume.

2.5. Zero queries dominate, and random splits are leaky. Since μ_G is computable classically, so is the entire depth-one landscape, and the number of quantum objective queries needed to find the best angle on any finite grid is zero. Any method that spends a positive budget is therefore dominated at

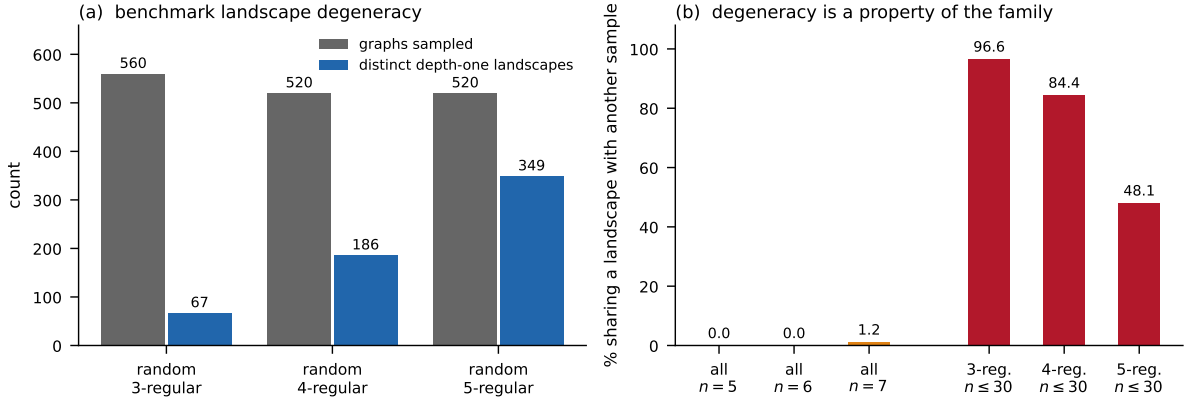


Fig. 3: Degeneracy is a property of the graph family. **a**, Random regular samples contain many fewer distinct depth-one landscapes than graphs. **b**, The fraction of graphs sharing a landscape with another sample is negligible for exhaustively enumerated small graphs but dominant for the regular families used as QAOA benchmarks.

depth one. We confirm this against three predictors trained on the corpus: the mean training landscape, ridge regression on graph features, and a ridge prior on the singular values of the adjacency matrix. Each ranks the angle grid, the target objective is queried in ranked order, and we report normalized regret after b queries. The exact landscape has zero regret at $b = 0$ by construction.

The more consequential finding concerns how such predictors are validated. Our corpus of 432 graphs contains only 231 distinct landscapes, so a split that randomizes over graphs does not randomize over problems. Under a uniform random split, 222 of 432 test graphs, or 51%, have a graph with a numerically identical landscape in their training set. Grouping the split by landscape instead removes the duplication, and the reported accuracies change qualitatively (Fig. 4). At one query the singular-value prior attains regret 0.00047 under the random split and 0.00551 under the grouped split, an inflation of roughly 12-fold. Under the random split it appears to be the best learned method, beating ridge at 0.00201; under the grouped split it is the worst, behind ridge at 0.00253. The training mean, which cannot exploit duplication because it ignores the input graph, is unchanged at 0.02599.

The apparent advantage of the singular-value prior is therefore an artifact of landscape duplication, not evidence of learned structure. This is the classical signature of leakage, and it is invisible to an audit based on graph isomorphism: across the 93,096 corpus pairs, 3,843 share a landscape exactly, of which 3,820 are isomorphic and 23 are not. An isomorphism-grouped protocol catches most but not all of the duplication; only grouping by the statistic itself removes it.

2.6. Exact landscapes at scale. Computing μ_G requires one pass for degrees and, per edge, an intersection of the smaller endpoint neighbourhood, giving $O(n + m + \sum_{(u,v) \in E} \min\{\deg u, \deg v\})$ preprocessing in $O(n + m)$ space. Evaluation then depends only on the number T of distinct triples, which is far smaller than m on real graphs; across every graph we timed, the median ratio m/T is 6.0. On the largest unweighted Gset instance, G22 with $n = 2,000$ and $m = 19,990$, the statistic has only 825 distinct triples, takes 0.07 s to build, and yields an exact 81-point landscape in 3 ms. The method reaches well past benchmark sizes: at $m = 1,199,424$ edges preprocessing takes 8.9 s and the landscape 1.0 s (Fig. 5).

The contrast with simulation is not close. At $n = 18$ vertices an 81-point landscape costs 6.1 s by statevector and 285 μ s from the statistic, a factor of about 21,200. The statevector cost grows as 2^n and the statistic cost does not grow with n at fixed m .

3. Discussion. The depth-one MaxCut landscape carries three integers per edge and nothing else. Once that is stated, the depth-one parameter transfer problem is solved rather than approximated: the landscape is free to compute, so there is nothing to transfer, and if one transfers anyway the loss is bounded in advance by a histogram distance. We verified that bound on 186,624 graph pairs without a violation.

The negative implications for evaluation are the part we expect to matter most. Benchmark suites built from random regular graphs contain far fewer distinct depth-one problems than graphs, and

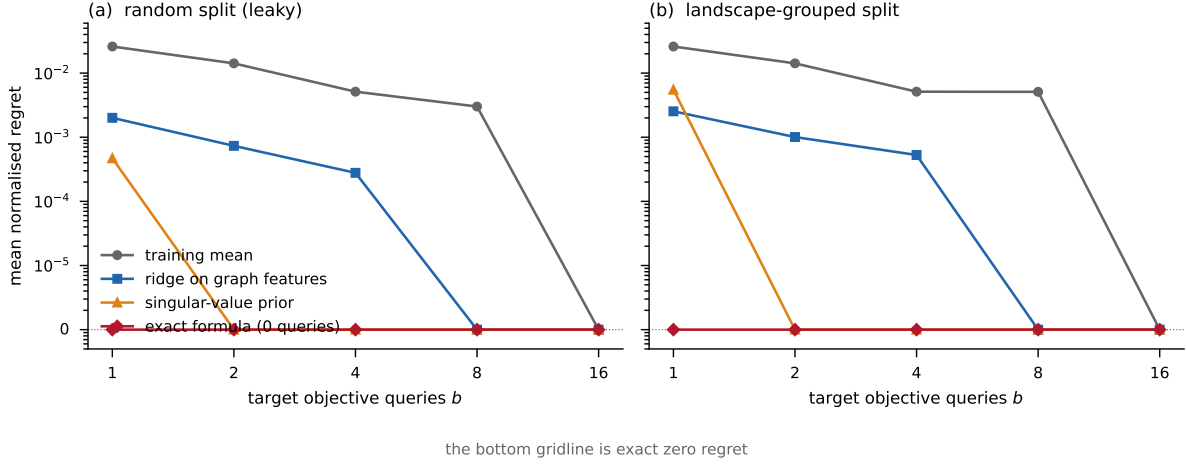


Fig. 4: Learned angle priors are dominated, and their measured accuracy depends on the split. Mean normalized regret against the number of target objective queries. **a**, A uniform random split over graphs, in which 51% of test graphs have an exact landscape duplicate in training. **b**, A split grouped by landscape. The singular-value prior is the best learned method in **a** and the worst in **b**. The exact landscape uses no queries and has identically zero regret in both.

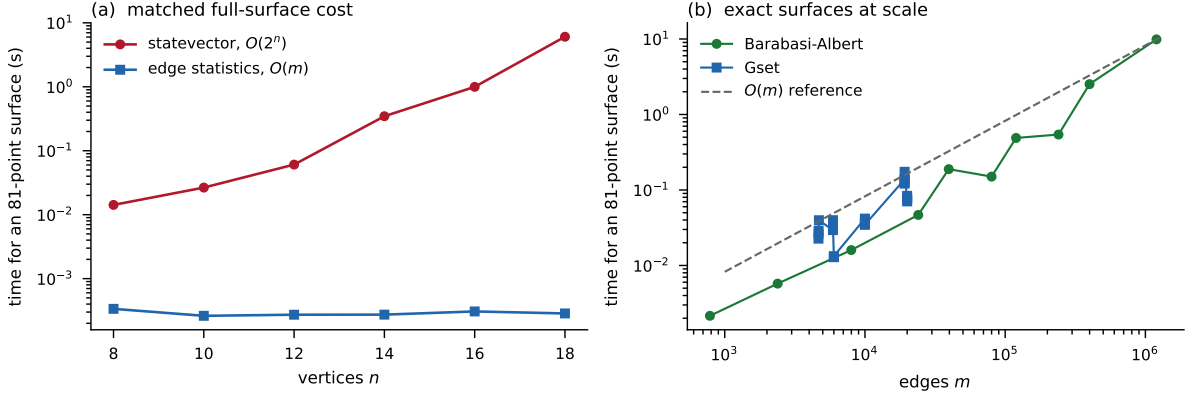


Fig. 5: Exact depth-one landscapes are cheap. **a**, Time for an 81-point landscape by statevector simulation and from the edge statistic, on cubic graphs. **b**, The same computation on Gset instances and on Barabási-Albert graphs up to 1,199,424 edges, against an $O(m)$ reference.

random train-test splits over such suites place exact duplicates on both sides. In our corpus this inflated the measured accuracy of a singular-value prior by an order of magnitude and reversed its ranking against a plain ridge baseline. Any depth-one study that reports a learned or transferred angle advantage should group its splits by the edge statistic and should compare against the exact landscape rather than against another learned method. Grouping by isomorphism is a partial remedy that misses genuine collisions, and grouping by graph identity is no remedy at all.

The scope is narrow and we do not stretch it. Everything above is depth one, unweighted and noiseless. The sufficient statistic is a property of the depth-one light cone, and at depth two the analogous object involves larger neighbourhoods and is not three-dimensional; whether a useful statistic of bounded dimension survives at $p \geq 2$, where closed forms are known only in restricted settings [7], is open and is the natural next question. Weighted objectives change f but not the shape of the argument, and we expect an analogous statistic on weighted triples, which we have not verified. Sampling noise and hardware error are not modelled here at all, and a finite-shot experiment does not inherit these guarantees. Finally, the fact that depth one is classically solved says nothing about the value of QAOA at depth where it is not; it says that depth one should no longer be used as evidence for query-efficiency claims.

4. Methods.

4.1. The depth-one edge identity.

Proposition 3 (Wang, Hadfield, Jiang and Rieffel [6]). *Let $\tau = (d_u, d_v, \lambda)$ be the triple of Eq. (2.2) for an edge (u, v) . Then the depth-one expectation of $C_{uv} = (I - Z_u Z_v)/2$ is $f(\tau; \gamma, \beta)$, where*

$$f = \frac{1}{2} + \frac{1}{4} \sin(4\beta) \sin \gamma (\cos^{d_u} \gamma + \cos^{d_v} \gamma) - \frac{1}{4} \sin^2(2\beta) \cos^{d_u+d_v-2\lambda} \gamma [1 - \cos^\lambda(2\gamma)]. \quad (4.1)$$

The derivation is that of Ref. [6] and we reproduce only its outline to fix notation. Conjugating by the mixer gives $e^{i\beta B} Z_u Z_v e^{-i\beta B} = c^2 Z_u Z_v + cs(Z_u Y_v + Y_u Z_v) + s^2 Y_u Y_v$ with $c = \cos(2\beta)$ and $s = \sin(2\beta)$. Cost terms outside the closed neighbourhood of (u, v) commute through and cancel. Conjugating the remaining Pauli strings by $e^{i\gamma C}$ and using $\langle +|P|+ \rangle = 0$ unless $P \in \{I, X\}^{\otimes n}$ leaves two endpoint contributions and one common-neighbour contribution, which are the three terms of Eq. (4.1).

4.2. Proof of Theorem 1. By Proposition 3 and linearity of the expectation in Eq. (2.1), $F_G = \frac{1}{m} \sum_{(u,v) \in E} f(\tau(u, v); \gamma, \beta)$, which is the average of f against the empirical distribution μ_G of the triples. Two graphs with the same μ_G therefore have the same F at every angle pair. Neither n nor the identity of any edge appears. $\square$

4.3. Proof of Theorem 2. C_{uv} is a projector, so its expectation in any state lies in $[0, 1]$ and f has oscillation $\text{osc } f = \sup f - \inf f \leq 1$. Let c be the midpoint of the range of f . Since μ_G and μ_H are probability measures, $\sum_\tau (\mu_G - \mu_H)(\tau) = 0$, and therefore

$$|F_G(\theta) - F_H(\theta)| = \left| \sum_\tau (\mu_G - \mu_H)(\tau) (f(\tau; \theta) - c) \right| \leq \frac{\text{osc } f}{2} \sum_\tau |(\mu_G - \mu_H)(\tau)|, \quad (4.2)$$

which is $\text{osc } f \cdot D_{\text{TV}}(\mu_G, \mu_H) \leq D_{\text{TV}}(\mu_G, \mu_H)$ and proves Eq. (2.4). For Eq. (2.5), write $\delta = D_{\text{TV}}(\mu_G, \mu_H)$ and decompose

$$\begin{aligned} F_H(\theta_H^*) - F_H(\theta_G^*) &= [F_H(\theta_H^*) - F_G(\theta_H^*)] + [F_G(\theta_H^*) - F_G(\theta_G^*)] \\ &\quad + [F_G(\theta_G^*) - F_H(\theta_G^*)]. \end{aligned} \quad (4.3)$$

The first and third brackets are at most δ by Eq. (2.4), and the second is at most zero because θ_G^* maximizes F_G over a set containing θ_H^* . $\square$

4.4. Numerical verification. Every claim that Eq. (4.1) is correct was checked against an independent statevector simulator written for this purpose, which builds the diagonal cost operator explicitly, applies the phase separator and the single-qubit mixer directly, and never uses the formula. Across 60 random graphs on four to fourteen vertices evaluated at 4,608 angle points each, a total of 276,480 values, the maximum absolute discrepancy is 1.25×10^{-14} . The collisions of Fig. 1 are reported from two statevector runs rather than from the formula, so that the agreement is not circular.

4.5. Corpus and protocols. The corpus of 432 graphs is deterministic given fixed seeds. It comprises ten-vertex breadth-first neighbourhoods of the first twelve unweighted Gset instances [8], taken from starts $37i \bmod n$, together with Erdős–Rényi graphs at three densities, random regular graphs of degree three and four, and Barabási–Albert graphs, each on ten, twelve and fourteen vertices. Of the Gset instances downloaded, 26 are unweighted and are the ones used; weighted instances are excluded because Eq. (4.1) does not apply to them.

Landscapes for the transfer study are evaluated on 4,608 angle points, with γ spanning its period $[0, 2\pi)$ and β spanning $[0, \pi/2)$. The query study uses the coarser nine-by-nine grid so that a small query budget is meaningful. Normalized regret after b queries is $r_b = (\max_\theta F - \max_{\theta \in Q_b} F) / \max_\theta F$, where Q_b holds the b grid points ranked highest by the predictor. This is a finite-grid statistic and asserts nothing about continuous global optimality. Predictors are ridge regressions with unit regularization on standardized features: order, size, density, the first four degree moments, triangle count and average clustering, together with the eight largest singular values of the adjacency matrix; the singular-value prior uses the spectral block alone. Splits are four-fold, either uniform over graphs or over landscape classes.

Data availability. The only external input is the Stanford Gset collection [8]. The package does not redistribute it; the fetch script records a SHA-256 checksum for each instance, and the pinned value for G1 is 73bf704d...e46d1665. All derived results are written to `code/results_full/results.json`.

Code availability. The implementation, the corpus builder, the three study scripts and the figure generator are in `code/`. Every number in this manuscript is emitted mechanically from `results.json` into `numbers.tex` by `code/scripts/make_numbers.py`; none is transcribed by hand.

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